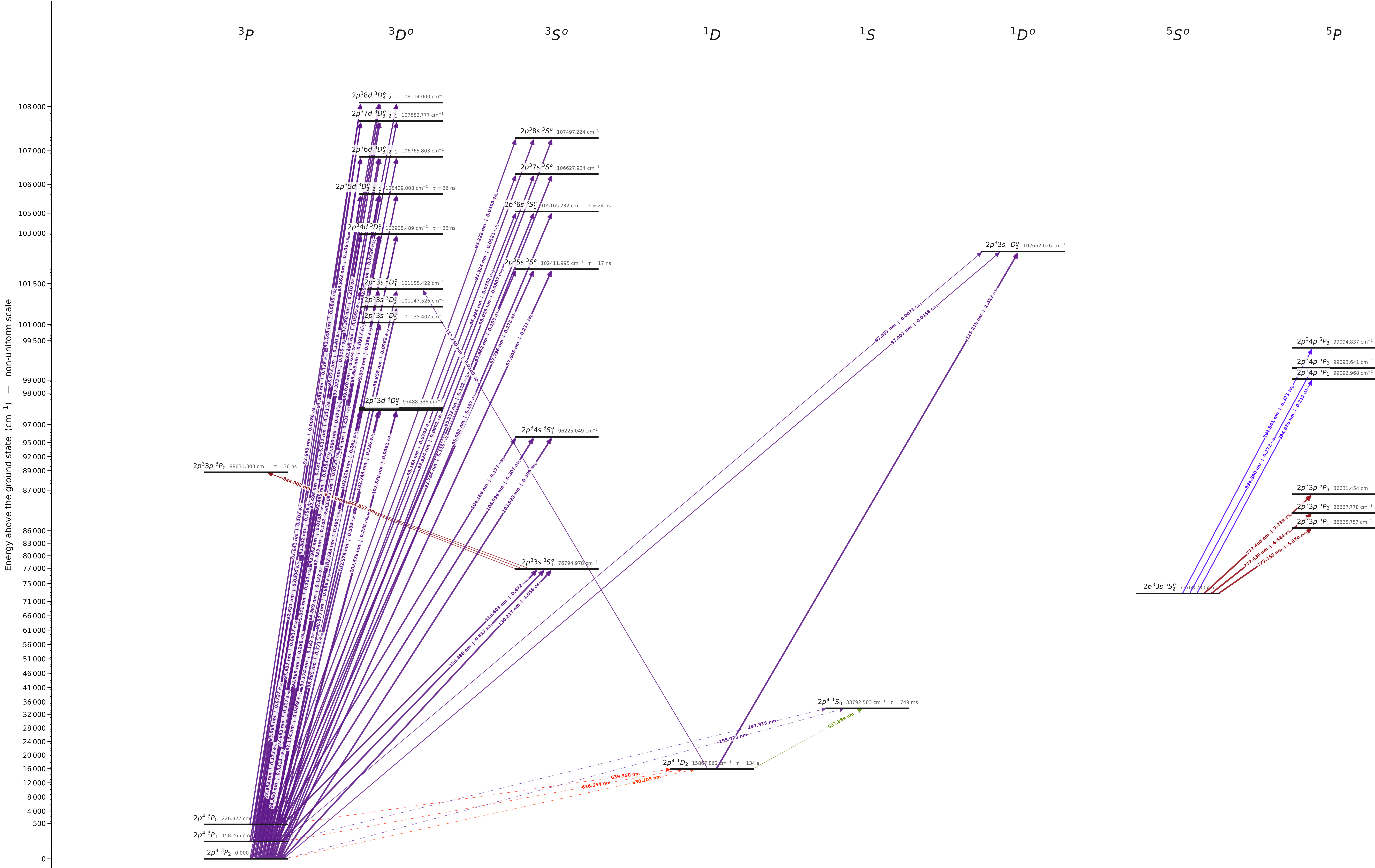
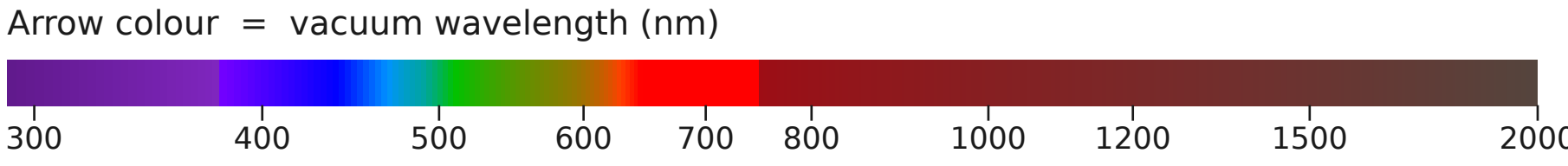


# Oxygen

O (O I). Strongest classified transitions below 2 μm (NIST): 43 levels, 79 lines.

O<sup>+</sup> ionisation limit 109837.02 cm<sup>-1</sup> = 13.61805 eV (91.044 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|\langle U||e\mathbf{r}||U'\rangle|$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                                                   | λ vac (nm) | λ air (nm) | ν (THz)   | ⟨U er U'⟩ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|--------------------------------------------------------------------------------------------------------------|------------|------------|-----------|------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 2p <sup>4</sup> <sup>3</sup> P <sub>1</sub> - 2p <sup>4</sup> <sup>1</sup> S <sub>0</sub>                    | 297.3154   | 297.2286   | 1008.3315 | -                            | 7.540e-02            | 2.126e-07      | 5.65       | ultraviolet auroral line [O I] 297.2 nm                      |
| 2p <sup>4</sup> <sup>1</sup> D <sub>2</sub> - 2p <sup>4</sup> <sup>1</sup> S <sub>0</sub>                    | 557.8887   | 557.7339   | 537.3696  | -                            | 1.260e+00            | 2.126e-07      | 94.3       | green auroral / airglow line [O I] 557.7 nm                  |
| 2p <sup>4</sup> <sup>3</sup> P <sub>2</sub> - 2p <sup>4</sup> <sup>1</sup> D <sub>2</sub>                    | 630.2046   | 630.0304   | 475.7065  | -                            | 5.630e-03            | 1.189e-09      | 75.3       | red auroral / airglow line [O I] 630.0 nm                    |
| 2p <sup>4</sup> <sup>3</sup> P <sub>1</sub> - 2p <sup>4</sup> <sup>1</sup> D <sub>2</sub>                    | 636.5536   | 636.3776   | 470.9619  | -                            | 1.820e-03            | 1.189e-09      | 24.3       | red auroral / airglow line [O I] 636.4 nm                    |
| 2p <sup>4</sup> <sup>3</sup> P <sub>0</sub> - 2p <sup>4</sup> <sup>1</sup> D <sub>2</sub>                    | 639.3500   | 639.1733   | 468.9019  | -                            | 8.600e-07            | 1.189e-09      | 0.01       | [O I] 639.2 nm                                               |
| 2p <sup>3</sup> 3s <sup>5</sup> S <sub>2</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>5</sup> P <sub>3</sub> | 777.4083   | 777.1944   | 385.6307  | 7.739                        | 3.690e+07            | -              | -          | 777.19 nm                                                    |
| 2p <sup>3</sup> 3s <sup>5</sup> S <sub>2</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>5</sup> P <sub>2</sub> | 777.6305   | 777.4166   | 385.5204  | 6.544                        | 3.690e+07            | -              | -          | 777.42 nm                                                    |
| 2p <sup>3</sup> 3s <sup>5</sup> S <sub>2</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>5</sup> P <sub>1</sub> | 777.7527   | 777.5388   | 385.4599  | 5.070                        | 3.690e+07            | -              | -          | 777.54 nm                                                    |
| 2p <sup>3</sup> 3s <sup>3</sup> S <sub>2</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>3</sup> P <sub>0</sub> | 844.8568   | 844.6247   | 354.8441  | -                            | -                    | 4.421          | -          | 844.6 nm fluorescence line of the O TALIF scheme: two-pho... |
| 2p <sup>3</sup> 3s <sup>3</sup> S <sub>1</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>3</sup> P <sub>2</sub> | 844.8680   | 844.6359   | 354.8394  | -                            | -                    | 4.421          | -          | 844.6 nm fluorescence line of the O TALIF scheme: two-pho... |
| 2p <sup>3</sup> 3s <sup>3</sup> S <sub>1</sub> <sup>o</sup> - 2p <sup>3</sup> 3p <sup>3</sup> P <sub>1</sub> | 844.9079   | 844.6758   | 354.8226  | -                            | -                    | 4.421          | -          | 844.6 nm fluorescence line of the O TALIF scheme: two-pho... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>17</sup>O hyperfine structure (I = 5/2)

| Level                                                       | A (MHz)     | B (MHz)   | Shift of each F level from the centre of gravity (MHz)                                   |
|-------------------------------------------------------------|-------------|-----------|------------------------------------------------------------------------------------------|
| 2p <sup>4</sup> <sup>3</sup> P <sub>2</sub>                 | -218.569(4) | -10.44(3) | F=1/2: +1522.68 F=3/2: +1199.52 F=5/2: +658.32 F=7/2: -104.85 F=9/2: -1095.45            |
| 2p <sup>4</sup> <sup>3</sup> P <sub>1</sub>                 | 4.74(4)     | 5.20(9)   | F=3/2: -12.94 F=5/2: -8.90 F=7/2: +13.14                                                 |
| 2p <sup>3</sup> 3s <sup>5</sup> S <sub>2</sub> <sup>o</sup> | -97.93(10)  | -         | F=1/2: +685.51 F=3/2: +538.62 F=5/2: +293.79 F=7/2: -48.97 F=9/2: -489.65                |
| 2p <sup>3</sup> 3p <sup>5</sup> P <sub>1</sub>              | -25.83(10)  | -         | F=3/2: +90.41 F=5/2: +25.83 F=7/2: -64.58                                                |
| 2p <sup>3</sup> 3p <sup>5</sup> P <sub>2</sub>              | -24.47(11)  | 0.9(8)    | F=1/2: +171.92 F=3/2: +134.81 F=5/2: +73.19 F=7/2: -12.62 F=9/2: -122.12                 |
| 2p <sup>3</sup> 3p <sup>5</sup> P <sub>3</sub>              | -18.70(5)   | -0.2(5)   | F=1/2: +186.88 F=3/2: +158.88 F=5/2: +112.20 F=7/2: +46.81 F=9/2: -37.34 F=11/2: -140.30 |

Measured A, B constants; sources in the data file.